

Hallucination Patterns in RAG-based Financial QA

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M1 Proposal ✓

M2 Preliminary paper ✓ submitted

M3 Final paper — in progress

RESEARCH QUESTIONS

- **RQ1** How often do RAG configurations hallucinate?
- **RQ2** Which hallucination **types** dominate in finance?
- **RQ3** Do larger / instruction-tuned models hallucinate less?

Why it's hard: answers are short & numeric, only **2%** appear verbatim in the evidence (they must be *computed*), and evidence is **69% tables** inside ~128-page filings — the regime where RAG hallucinates.

WHAT WE'VE BUILT

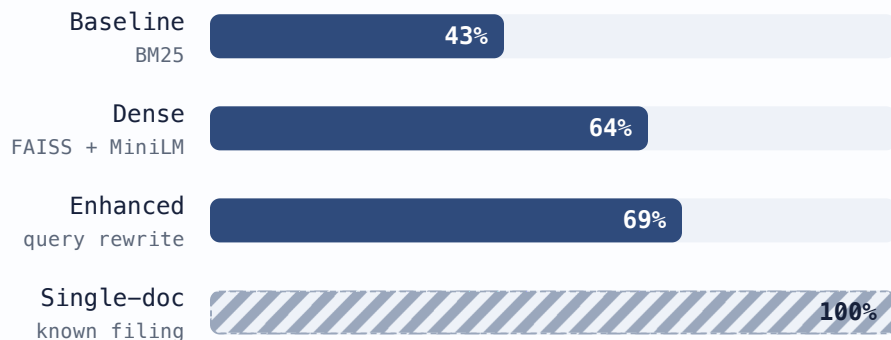
- **Data & EDA** — 150 QA over 84 filings; 22-figure exploratory analysis
- **3 RAG pipelines** — BM25, dense (FAISS+MiniLM), query-rewrite — all on GPT-3.5
- **Evaluation harness** — RAGAS (faithfulness / relevancy / precision) + F1 / EM
- **Reusable design** — one output schema scores every pipeline unchanged
- **Related work** — 24 papers across RAG, hallucination, financial NLP

RESULTS & NEXT STEPS

Retrieval — not the generator — is the dominant lever

Same generator, prompt, chunks & top-3 across all four; only **retrieval** changes (all 150 questions, GPT-3.5, temp 0).

CORRECT FILING FOUND IN TOP-3 (RETR@3)



■ Single-doc = upper-bound ablation (retrieval restricted to the correct filing), not a deployable pipeline.

EVERY METRIC RISES WITH RETRIEVAL

CONFIG	FAITH*	CTX.PREC	ANSW.	F1	EM#
Baseline	0.41	0.24	47	0.083	0.12
Dense	0.35	0.42	82	0.116	0.18
Enhanced	0.35	0.46	87	0.121	0.21
Single-doc	0.57	0.44	96	0.139	0.27

RAGAS judge = GPT-4o-mini. Faith* = faithfulness on *answered* questions; EM# = numeric-tolerant EM (strict EM ≈ 0 — formatting, not errors). Answ. = answered of 150.

Takeaway: dense retrieval + query rewriting lift coverage 47 → 87 of 150, context precision, F1, and numeric-EM (0.12 → 0.21) together — yet even the 100%-retrieval ceiling leaves a third unanswered and answered-faithfulness at 0.57, so **finding the filing is necessary but not sufficient**. Those grounded-but-wrong cases are the material for the taxonomy.

NEXT

T16 chunk / top-k sweeps

T17 Mistral-7B — RQ3

T19–23 annotate 50 failures → 4-type taxonomy — RQ2

T27 final 8-page paper